

Coronary Artery Disease (Ischemic Heart Disease and Myocardial Infarction)

- It is a condition in which there is an inadequate supply of blood and oxygen to a portion of the myocardium
- It typically occurs when there is an imbalance between myocardial oxygen supply and demand.
- The most common cause of myocardial ischemia is atherosclerotic disease of an epicardial coronary artery (or arteries) sufficient to cause a regional reduction in myocardial blood flow and inadequate perfusion of the myocardium supplied by the involved coronary artery.

Epidemiology and risk factors

- IHD causes more deaths and disability and incurs greater economic costs than any other illness in the developed world.
- Genetic factors, a high-fat and energy-rich diet, smoking, and a sedentary lifestyle are associated with the emergence of IHD
- Obesity, insulin resistance, and type 2 diabetes mellitus are increasing and are powerful risk factors for IHD.
- With urbanization in countries with emerging economies and a growing middle class, elements of the energy-rich Western diet are being adopted.
- In light of the projection of large increases in IHD throughout the world, IHD is likely to become the most common cause of death worldwide by 2020.

Pathophysiology

- Central to an understanding of the pathophysiology of myocardial ischemia is the concept of myocardial supply and demand
- By reducing the lumen of the coronary arteries,

atherosclerosis limits appropriate increases in perfusion when the demand for flow is augmented, as occurs during exertion or excitement.

- When the luminal reduction is severe, myocardial perfusion in the basal state is reduced.
- Coronary blood flow also can be limited by spasm (see Prinzmetal's angina) arterial thrombi, and, rarely, coronary emboli as well as by ostial narrowing due to aortitis.
- Congenital abnormalities such as the origin of the left anterior descending coronary artery from the pulmonary artery may cause myocardial ischemia and infarction in infancy; but this cause is very rare in adults.
- Myocardial ischemia also can occur if myocardial oxygen demands are markedly increased and particularly when coronary blood flow may be limited, as occurs in severe left ventricular hypertrophy due to aortic stenosis. The latter can present with angina that is indistinguishable from that caused by coronary atherosclerosis largely owing to subendocardial ischemia
- A reduction in the oxygen-carrying capacity of the blood, as in extremely severe anemia or in the presence of carboxyhemoglobin, rarely causes myocardial ischemia by itself but may lower the threshold for ischemia in patients with moderate coronary obstruction.

Microvascular angina

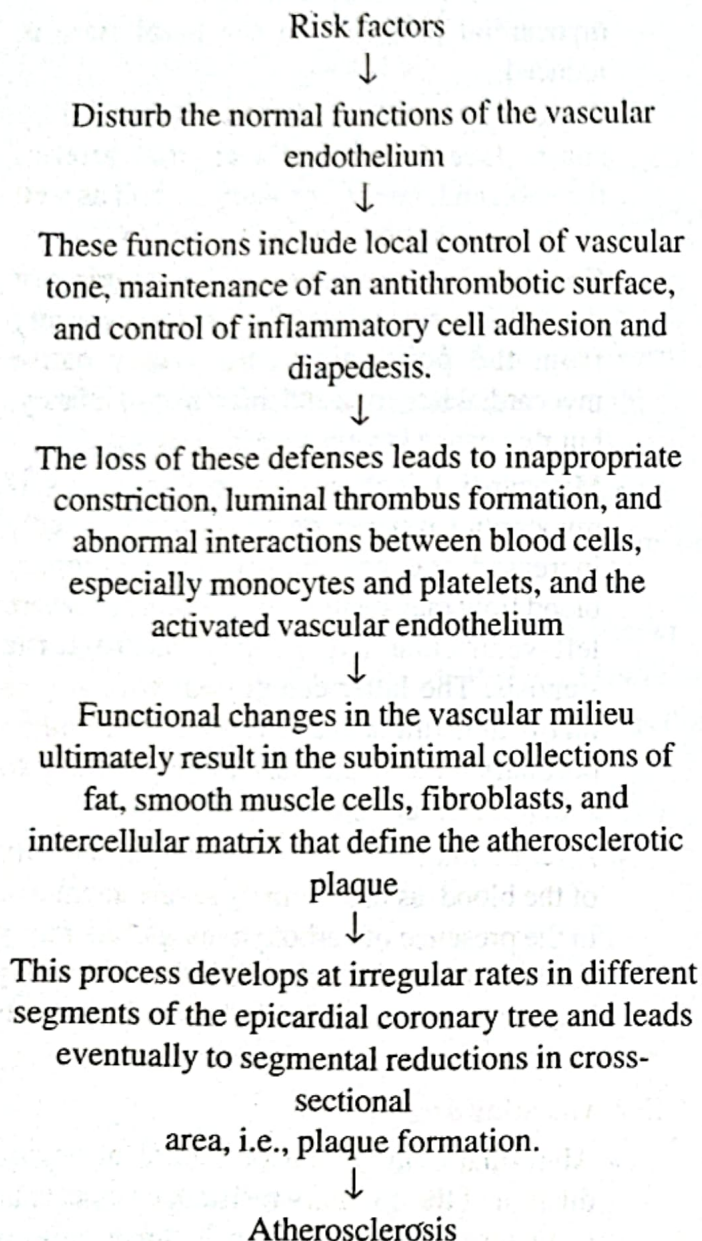
- Abnormal constriction or failure of normal dilation of the coronary resistance vessels also can cause ischemia. When it causes angina, this condition is referred to as *microvascular angina*.

Coronary Atherosclerosis

- Epicardial coronary arteries are the major site of atherosclerotic disease.

Major risk factors

- High levels of plasma low-density lipoprotein (LDL)
- Low plasma high-density lipoprotein (HDL)
- Cigarette smoking
- Hypertension
- Diabetes mellitus

Pathogenesis**Stable Angina Pectoris**

- Transient myocardial ischemia causes episodic clinical syndrome
- Males constitute 70% of all patients with angina pectoris and an even greater proportion of those less than 50 years of age.

- It is, however, important to note that angina pectoris in women is often atypical in presentation

History-clinical features

- The typical patient with angina is a man >50 years or a woman >60 years of age who complains of episodes of chest discomfort, usually described as heaviness, pressure, squeezing, smothering, or choking and only rarely as frank pain.

Levine's sign

- When the patient is asked to localize the sensation, he or she typically places a hand over the sternum, sometimes with a clenched fist, to indicate a squeezing, central, substernal discomfort
- Angina is usually crescendo-decrescendo in nature, typically lasts 2 to 5 min, and can radiate to either shoulder and to both arms (especially the ulnar surfaces of the forearm and hand).
- It also can arise in or radiate to the back, interscapular region, root of the neck, jaw, teeth, and epigastrium.
- Angina is rarely localized below the umbilicus or above the mandible.
- A useful finding in assessing a patient with chest discomfort is the fact that myocardial ischemic discomfort does not radiate to the trapezius muscles; that radiation pattern is more typical of pericarditis.
- Although episodes of angina typically are caused by exertion (e.g., exercise, hurrying, or sexual activity) or emotion (e.g., stress, anger, fright, or frustration) and are relieved by rest
- Pain may occur at rest- "Unstable Angina Pectoris,"
- While the patient is recumbent causes pain-Angina Decubitus.

Nocturnal angina

- The patient may be awakened at night by typical chest discomfort and dyspnea.
- Nocturnal angina may be due to episodic

tachycardia, diminished oxygenation as the respiratory pattern changes during sleep, or expansion of the intrathoracic blood volume that occurs with recumbency; the latter causes an increase in cardiac size (end-diastolic volume), wall tension, and myocardial oxygen demand that can lead to ischemia and transient left ventricular failure.

- The threshold for the development of angina pectoris may vary by time of day and emotional state.

Physical Examination

- Abdominal aortic aneurysm
- Carotid arterial bruits
- Diminished arterial pulses in the lower extremities.
- Xanthelasmas
- Xanthomas
- Pulse contour at multiple locations
- Comparing the blood pressure between the arms and between the arms and the legs (ankle-brachial index).
- Examination of the fundi may reveal an increased light reflex and arteriovenous nicking as evidence of hypertension.
- There also may be signs of anemia, thyroid disease, and nicotine stains on the fingertips from cigarette smoking.
- A protuberant abdomen may indicate that the patient has the metabolic syndrome and is at increased risk for atherosclerosis.

Palpation

- It may reveal cardiac enlargement and abnormal contraction of the cardiac impulse (left ventricular dyskinesia).
- Tenderness of the chest wall, localization of the discomfort with a single fingertip on the chest, or reproduction of the pain with palpation of the chest makes it unlikely that the pain is caused by myocardial ischemia.

Auscultation

- These auscultatory signs are best appreciated

with the patient in the left lateral decubitus position.

- Auscultation can uncover arterial bruits, a third and/or fourth heart sound
- Apical systolic murmur due to mitral regurgitation.

Laboratory Examination

- Microalbuminuria -The urine should be examined for evidence of diabetes mellitus and renal disease

Examination of the blood

- Measurements of lipids (cholesterol—Total, LDL, HDL—and triglycerides)
- Glucose (hemoglobin A_{1c})
- Creatinine
- Hematocrit
- Thyroid function.

A chest x-ray

- A chest x-ray is important as it may show the consequences of IHD, i.e., cardiac enlargement, ventricular aneurysm, or signs of heart failure.

C-reactive protein (CRP)

- Evidence exists that an elevated level of high-sensitivity C-reactive protein (CRP) (specifically, between 0 and 3 mg/dL) is an independent risk factor for IHD

The major benefit of high-sensitivity CRP is in reclassifying the risk of IHD in patients in the “intermediate” risk category on the basis of traditional risk factors

Congestive Cardiac failure

Congestive heart failure (CHF) is a common clinical syndrome, and conventionally implies right ventricular failure associated with left heart failure. However, the latter may be minimal or sometimes even absent as in chronic cor-pulmonale. It is usually a chronic condition but may occasionally be acute as with massive myocardial infarction, or following an abrupt increase in pre-existing heart failure. It may then become a medical emergency requiring prompt treatment.

Etiopathology

Nearly all cases of CHF have an obvious underlying disease involving the myocardium or the cardiac valves, the most frequent being rheumatic, ischemic or hypertensive heart disease, thyrotoxicosis, various forms of carditis and cardiomyopathy, and chronic cor-pulmonale. It is also associated with concomitant infection, acute massive hemorrhage, after surgical operation or from discontinuation of digitalis and diuretics. It is important to identify such precipitating causes both for immediate management as well as to prevent acute recurrence.

The pathophysiology of CHF is complex. It is certainly multifactorial, the two most significant factors being: (i) a decrease in cardiac output due to either impaired cardiac contraction or insufficient cardiac filling; and (ii) consequent adaptive measures of which activation of neuro-hormonal systems is specially important. This includes release of norepinephrine and activation of renin-angiotensin-aldosterone system. These adaptive mechanisms increase heart rate, blood and cardiac filling pressure, and cause redistribution of blood flow an attempt to maintain adequate overall pumping functions of the heart, and preserve perfusion of the vital organs. Frank heart failure occurs when the capacity of these adaptive mechanisms is superseded, and, depending upon the rapidity and severity the cardiac performance becomes impaired.

Clinical features

1. Fatigability and varying degree of breathlessness
2. Orthopnoea
3. Paroxysmal nocturnal dyspnea
4. Mild cyanosis
5. Tachycardia
6. Irregular pulse
7. Pulmonary rales- from the base upwards with the worsening of LHF.

O/E : Advanced cases invariably develop the classical triad of right heart failure viz., elevated

jugular venous pressure, tender hepatomegaly and subcutaneous edema over dependent parts of the body. More severe cases may develop serous effusions and extensive edema involving the whole body. The blood pressure is usually low with a small pulse pressure.

MCQs

1. Abdominal aortic aneurysm, carotid arterial bruits, and diminished arterial pulses in the lower extremities indicates
 - a. Atherosclerotic disease
 - b. Aortic aneurysm
 - c. Diabetes mellitus
 - d. Peripheral arterial disease
2. Homan's sign is neither specific nor sensitive for diagnosis of
 - a. Rheumatoid arthritis
 - b. Muscular atrophy
 - c. Primary neuromuscular disorder.
 - d. Deep venous thrombosis
3. ————usually develops gradually with exertion, emotional excitement, or after heavy meals.
 - a. Stable angina pectoris
 - b. Unstable Angina.
 - c. Myocardial Infarction
 - d. Pericarditis
4. Which of the following statements is CORRECT?
 - a. Myocardial tissue necrosis occurs immediately when blood flow is occluded.
 - b. Cardiac troponin is highest 30 minutes after infarction.
 - c. Inflammatory biomarkers are often elevated after an acute MI.
 - d. Chest pain is always present in myocardial infarctions.

Answer

1. a 2. d 3. a 4. c